

# Problems In Group Theory

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# Chapter 1

## Basic Group Theory

### 1.1 Introduction

**Problem 1.1.1.** Determine which of the following binary operations are associative:

1. the operation on  $\mathbb{Z}$  defined by  $a * b = a - b$
2. the operation on  $\mathbb{R}$  defined by  $a * b = a + b + ab$
3. the operation on  $\mathbb{Q}$  defined by  $a * b = \frac{a+b}{5}$
4. the operation on  $\mathbb{Z} \times \mathbb{Z}$  defined by  $(a, b) * (c, d) = (ad + bc, bd)$
5. the operation on  $\mathbb{Q} - \{0\}$  defined by  $a * b = \frac{a}{b}$

*Proof.* This exercise is pretty trivial as we just have to compute  $a * (b * c) = (a * b) * c$ . For 1.,  $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$  but  $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$ . Rest of them are pretty similar.  $\square$

**Problem 1.1.2.** Prove that addition of residue classes in  $\mathbb{Z}/n\mathbb{Z}$  is associative.

*Proof.* By definition of residue addition,

$$[a] + [b] = [a + b]$$

Therefore, we get the following.

$$\begin{aligned} [a] + ([b] + [c]) &= [a] + [b + c] \\ &= [a + (b + c)] \\ &= [(a + b) + c] \\ &= [a + b] + [c] \\ &= ([a] + [b]) + [c] \end{aligned}$$

$\square$

**Problem 1.1.3.** Prove that multiplication of residue classes in  $\mathbb{Z}/n\mathbb{Z}$  is associative.

*Proof.* By definition of residue multiplication,

$$[a] \cdot [b] = [a \cdot b]$$

Therefore, we get the following.

$$\begin{aligned} [a] \cdot ([b] \cdot [c]) &= [a] \cdot [b \cdot c] \\ &= [a \cdot (b \cdot c)] \\ &= [(a \cdot b) \cdot c] \\ &= [a \cdot b] \cdot [c] \\ &= ([a] \cdot [b]) \cdot [c] \end{aligned}$$

□

**Problem 1.1.4.** *Prove for all  $n > 1$  that  $\mathbb{Z}/n\mathbb{Z}$  is not a group under multiplication of residue classes.*

*Proof.* There is no multiplicative inverse for  $[a]$  with  $\gcd(a, n) > 1$ . The proof for it pretty trivial, if there was a inverse, it would mean that

$$a \cdot b \equiv 1 \pmod{n} \implies 0 \equiv 1 \pmod{\gcd(a, n)}$$

That's a contradiction.

□

**Problem 1.1.5.** *Determine which of the following sets are groups under addition:*

1. *the set of rational numbers (including  $0 = 0/1$ ) in lowest terms whose denominators are odd*
2. *the set of rational numbers (including  $0 = 0/1$ ) in lowest terms whose denominators are even*
3. *the set of rational numbers of absolute value  $< 1$*
4. *the set of rational numbers of absolute value  $\geq 1$  together with 0*
5. *the set of rational numbers with denominators equal to 1 or 2*
6. *the set of rational numbers with denominators equal to 1, 2 or 3.*

*Proof.* Below are the answers:

1. Group.
2. Not a group. Because it is not closed. (Eg:  $5/2 + 7/2 = 6$ )
3. Not a group. Because again it is not closed. (Eg :  $0.99 + 0.98 = 1.97$ )
4. Not a group. Because again it is not closed. (Eg :  $4 + (-3.1) = 0.9$ )
5. Group.
6. Not a group. Because its not closed. (Eg :  $1/2 + 1/3 = 5/6$ )

□

**Problem 1.1.6.** Let  $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$  and for  $x, y \in G$  let  $x * y$  be the fractional part of  $x + y$  (i.e.,  $x * y = x + y - \lfloor x + y \rfloor$  where  $\lfloor a \rfloor$  is the greatest integer less than or equal to  $a$ ). Prove that  $*$  is a well defined binary operation on  $G$  and that  $G$  is an abelian group under  $*$  (called the real numbers mod 1).

*Proof.* It is well defined as,

$$\begin{aligned} x_1 = x_2, y_1 = y_2 &\implies x_1 * y_1 = x_1 + y_1 - \lfloor x_1 + y_1 \rfloor \\ &\implies x_1 + y_1 - \lfloor x_1 + y_1 \rfloor = x_2 + y_2 - \lfloor x_2 + y_2 \rfloor = x_2 * y_2 \end{aligned}$$

Also, its closed as  $x * y = \{x + y\}$  and  $0 \leq x + y < 1$ . One can check that inverses and identity exists. For associativity,

$$\begin{aligned} x * (y * z) &= x * (y + z - \lfloor y + z \rfloor) \\ &= \{x + y + z - \lfloor y + z \rfloor\} \\ &= \{x + y + z\} \\ &= \{x + y - \lfloor x + y \rfloor + z\} \\ &= (x * y) * z \end{aligned}$$

□

**Problem 1.1.7.** Let  $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$ .

1. Prove that  $G$  is a group under multiplication (called the group of roots of unity in  $\mathbb{C}$ ).
2. Prove that  $G$  is not a group under addition.

*Proof.* For  $G$  under multiplication, we know that for all  $z_1, z_2, z_3 \in G$  we have

$$z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$$

Also, it is closed as for all  $z_1, z_2 \in G$  we have  $z_1^n = 1$  and  $z_2^k = 1$  so

$$(z_1 \cdot z_2)^{\text{lcm}(n,k)} = 1$$

Now, for inverse of a  $z \in G$  just take  $1/z$ . As  $(1/z)^n = 1/1 = 1$  so  $1/z \in G$ . Also,  $z \cdot 1/z = 1$ .

For  $G$  under addition, if it was a group then,  $1 \in G$  as  $1^1 = 1 \in G \implies 1 + 1 = 2 \in G$  but there does not exist any integer  $n$  such that  $2^n = 1$ . □

**Problem 1.1.8.** Let  $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$ .

1. Prove that  $G$  is a group under addition.
2. Prove that the nonzero elements of  $G$  are a group under multiplication.

*Proof.* We know reals are associative so the elements of  $G$  are associative under addition. The inverse for  $a + b\sqrt{2}$  is  $(-a) + (-b)\sqrt{2}$ . It is also closed as,

$$a + b\sqrt{2} + c + d\sqrt{2} = (a + c) + (b + d)\sqrt{2}$$

The identity is going to be  $0 + 0 \cdot \sqrt{2} = 0$ .

We know reals are associative so the elements of  $G$  are associative under multiplication. The inverse for  $a + b\sqrt{2}$  is  $\frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2}$ . The identity is  $1 + 0 \cdot \sqrt{2} = 1$ . And its closed under multiplication as

$$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$$

□

**Problem 1.1.9.** Prove that a finite group is abelian if and only if its group table is a symmetric matrix.

*Proof.* Let  $G = \{g_1, \dots, g_n\}$  be a group with  $n$  elements. Then we define the group matrix to be a  $n \times n$  square matrix with entries  $a_{ij} = g_i g_j$ . So, if its  $G$  is abelian then  $a_{ij} = g_i g_j = g_j g_i = a_{ji}$ , so its symmetric. If the group table is symmetric then  $a_{ij} = a_{ji} \implies g_i g_j = g_j g_i$  for all  $1 \leq i, j \leq n$ . Thus, its abelian. □

**Problem 1.1.10.** Find the orders of each element of the additive group  $\mathbb{Z}/12\mathbb{Z}$ .

*Proof.* For  $[1], [5], [7], [11]$  its going to be 12. For  $[2], [10]$  its going to be 6, for  $[4], [8]$  its going to be 3, for  $[6]$  its going to be 2. For  $[3], [9]$  its going to be 4. For  $[0]$  its going to be 1. □

**Problem 1.1.11.** Prove that  $(a_1 a_2 \dots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}$  for all  $a_1, a_2, \dots, a_n \in G$ .

*Proof.* We will use strong induction. Its obviously true for  $n = 1$  and assume its true for  $n = 2, \dots, k$ . Then,

$$\begin{aligned} ((a_1 \dots a_k) \cdot a_{k+1})^{-1} &= a_{k+1}^{-1} (a_1 \dots a_k)^{-1} \\ &= a_{k+1}^{-1} a_k^{-1} \dots a_1^{-1} \end{aligned}$$

□

**Problem 1.1.12.** Let  $x$  be an element of  $G$ . Prove that  $x^2 = 1$  if and only if  $|x|$  is either 1 or 2.

*Proof.* Since  $x^2 = 1$  we have  $|x| \leq 2$ , and since  $|x|$  is a positive integer we have  $|x| = 1, 2$ . And if  $|x| = 1$  then  $x = 1 \implies x^2 = x = 1$  and if  $|x| = 2$  then  $x^2 = 1$ . □

**Problem 1.1.13.** Let  $x$  be an element of  $G$ . Prove that if  $|x| = n$  for some positive integer  $n$  then  $x^{-1} = x^{n-1}$ .

*Proof.* If  $|x| = n$  then  $x^n = 1 \implies x^{n-1} * x^1 = 1 \implies x^{n-1} = 1 * x^{-1} = x^{-1}$ . □